Impact Metrics

Team: Questers

Prachi Mittal

EXECUTIVE SUMMARY : AQI Risk Concentration

High-level problem overview with city-wise and station-level impact assessment

Annual Socio-Awareness Fest, IIT (BHU) VARANASI
JAGRITI'26
6th FEB - 8th FEB

Key Metrics

5+ Years
Historical AQI data

26+ Cities
Pan-India Coverage

120+ Stations
Active Monitoring Network

1M+ Records
Hourly and Daily Observations

PM2.5 & PM10
Primary AQI drivers

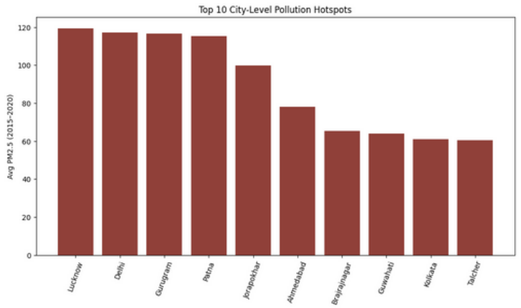
70%+ Cities
Above Safe Limits

Problem Overview

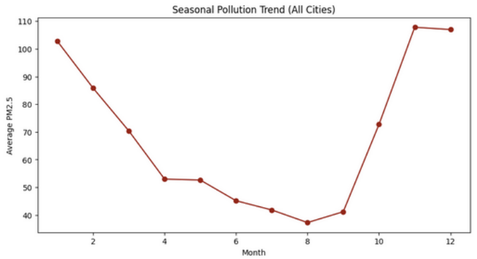
current situation of air quality in a city

Reality

- **Top 10 cities** account for disproportionately **high PM2.5**
- **NCR dominates** station-level extremes
- **Pollution is persistent**, not episodic

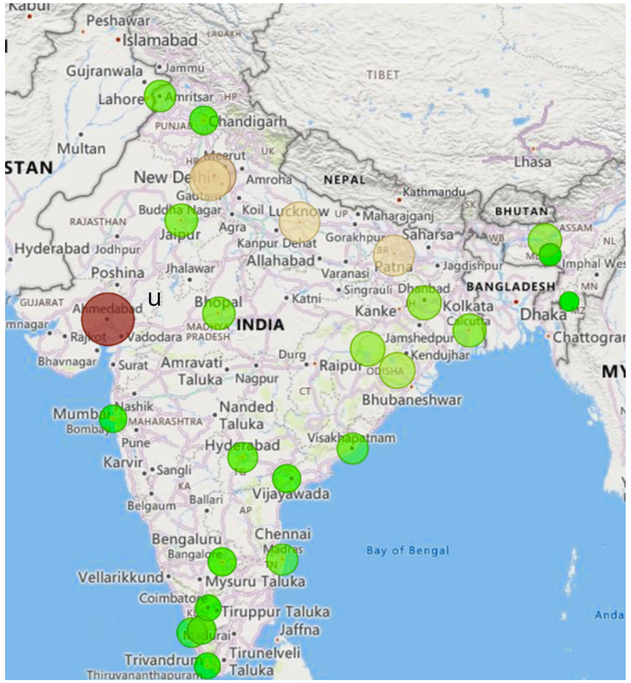


- **City averages** mask station hotspots
- **Seasonal events** sharply amplify exposure



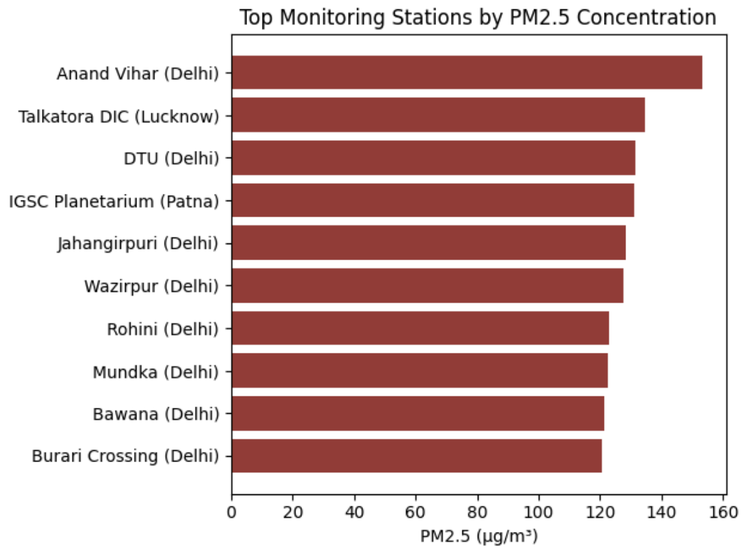
Pain

Air Pollution is Highly Clustered Across Cities



- **70%+ cities** exceed WHO safe PM2.5 limits
- Top clusters record **PM2.5 >120 µg/m³**
- NCR contributes majority of **extreme AQI hotspots**
- Southern/coastal cities remain **<60 µg/m³** on average

Station-Level Hotspots



- Station-level PM2.5 peaks **exceed city averages by 20–30%.**
- **Multiple hotspots exist** within the same city (NCR dominance).
- **Pollution intensity** varies sharply within cities.

Highest station-level PM2.5 observed — **153 µg/m³**

Situation Overview | Key Drivers | Impact and it's Action

Code Link

Drivers of Air Pollution Peaks

AQI spikes are driven by winter inversion, crop-burning events, and dominant emission sources

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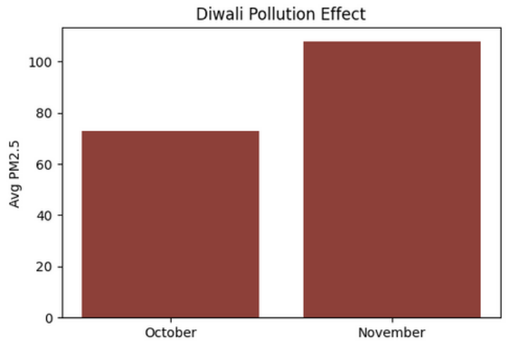
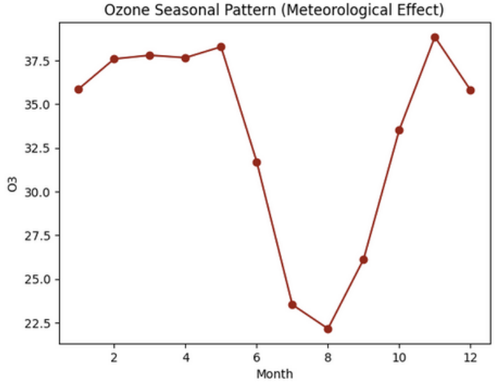
Seasonal & Event-Driven Effects



Diwali Effect (PM2.5)

- **October**: ~70 µg/m³
- **November**: ~108 µg/m³
- ~50% jump post-festival

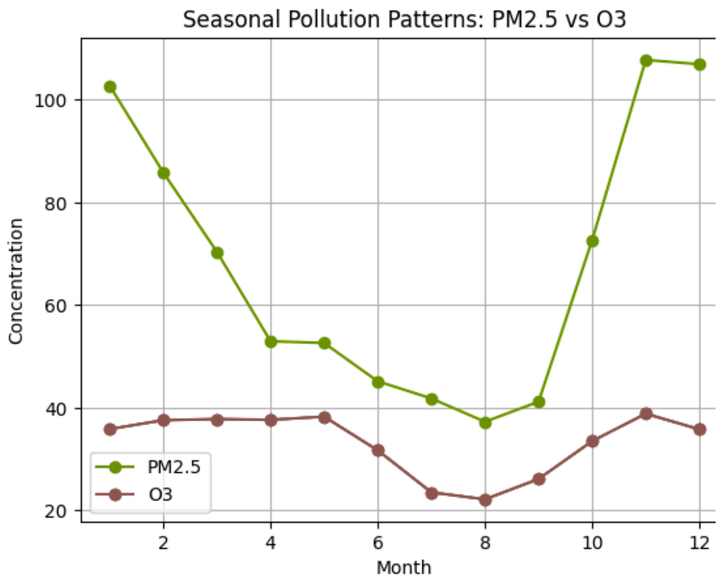


Ozone Seasonal Pattern

- Apr–Jun peak → sunlight chemistry
- Jul–Sep dip → monsoon washout

Driver	Control Focus
Winter Inversion	Dust suppression, construction bans
Crop burning	Regional coordination
Vehicular NO_2 / CO	Traffic restrictions, fuel shift
Summer O_3	Precursor emission control

Seasonal Pollution Patterns



Nov - Jan	April- June	July - Sept
▲PM2.5 peak	▲O_3 peak	▼ lowest PM2.5
Winter Inversion Spike	Sunlight-Driven O_3 Formation	Monsoon Washout
• Low wind + temperature inversion • PM2.5 accumulation	• High temperature •Photochemical reactions dominate	• Rain removes particulates • Cleanest air period

Pollution Source Signals

PM2.5 ↔ PM10 (High Correlation)

- Indicates shared sources such as road dust and combustion emissions
- Dominant contributors to overall AQI across urban cities

NO_2 ↔ CO (High Correlation)

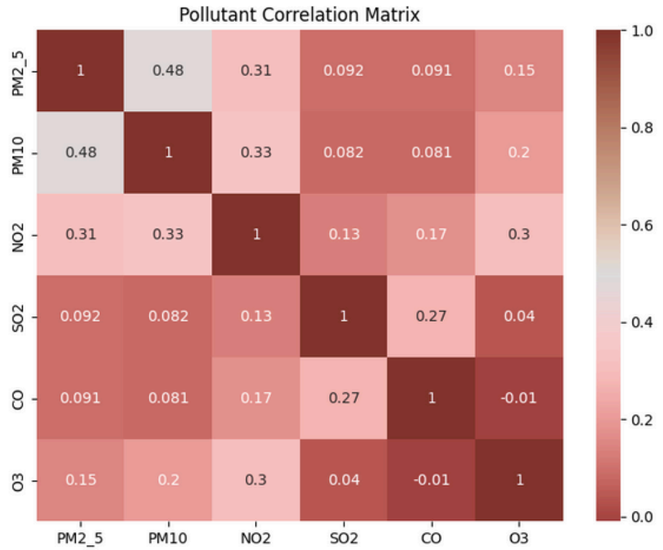
- Strongly linked to vehicular exhaust and traffic density
- Prominent near highways and high-mobility zones

SO_2 (Weak Correlation)

- Suggests localized industrial point sources
- Limited influence on city-wide AQI trends

O_3 ↔ PM (Weak / Negative Correlation)

- Driven by sunlight and meteorological conditions
- Peaks in summer despite lower particulate levels



Impact & Action: Translating Insights into Decisions

Who is most at risk and what must be done now

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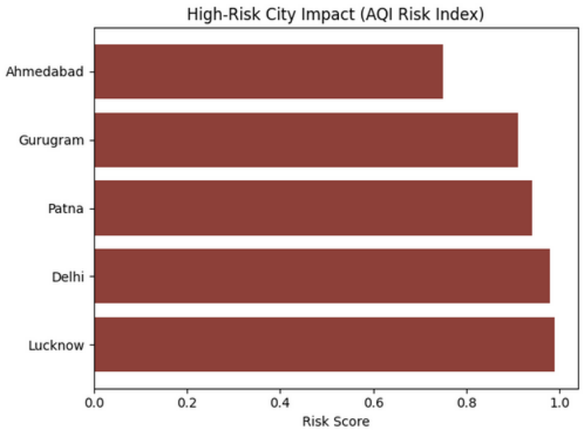
High-Risk City Impact



- **NCR & Indo-Gangetic** cities dominate risk rankings
- **Long exposure × high PM2.5** → compounded health burden

Vulnerability Lens

Children → Lung Development risk

Elderly → Cardio-respiratory stress

Outdoor Workers → chronic exposure

Targeted Action Framework

Problem Driver	Action Lever
Winter PM spike	Seasonal construction & dust controls
Vehicular emissions	Traffic restrictions, fuel shift
Crop burning	Regional coordination & incentives

Action Prioritization Rule

- High Risk Score + Long Exposure → **Immediate Intervention**
- Short-term spikes → **Event-based controls**
- Data blind spots → **Monitoring investment**

Air pollution in India is predictable, local, and solvable with targeted action.

KPI Strips

Top Risk Cities: **Lucknow | Delhi | Patna | Gurugram**

Exposure Duration: **1,600–1,950 high-pollution days**

Peak Winter PM2.5: **~200 µg/m³**

Monitoring Gaps: **30–50% station-year coverage missing**

Thank you